MMF 1921 Homework Assignment 1 2026

Due May 21 by 11:59PM. Submit through quercus.. You must write up the formulation for each problem and show results of solving the model using tables or graphs with reasonable formatting (please do not just give the dump of the computational output from MATLAB, this as mentioned should go in an appendix).

Problem 1

Currently we own 100 shares each of stocks 1 through 10. The original price we paid for these stocks, today's price, and the expected price in one year for each stock is shown in Table 76.

We need money today and are going to sell some of our stocks. The tax rate on capital gains is 30%. If we sell 50 shares of stock 1, then we must pay tax of $.3 \cdot 50(30 - 20) = \$150$. We must also pay transaction costs of 1% on each transaction. Thus, our sale of 50 shares of stock 1 would incur transaction costs of $.01 \cdot 50 \cdot 30 = \$15$. After taxes and transaction costs, we must be left with \$30,000 from our stock sales. Our goal is to maximize the expected (before-tax) value in one year of our remaining stock. What stocks should we sell? Assume it is all right to sell a fractional share of stock.

TABLE 76

Stock	Shares Owned	Price ($)		
		Purchase	Current	In One Year
1	100	20	30	36
2	100	25	34	39
3	100	30	43	42
4	100	35	47	45
5	100	40	49	51
6	100	45	53	55
7	100	50	60	63
8	100	55	62	64
9	100	60	64	66
10	100	65	66	70
Tax rate (%)	0.3			
Transaction cost (%)	0.01			

Formulate the problem as a linear programming problem and solve using MATLAB. Define all variables and explain all constraints.

Please highlight the optimal decisions and objective function clearly in your report.

Read Chapter 1 in the text and do the following exercises:

Problem 2 Exercise 1.3
Problem 3 Exercise 1.8
Problem 4 Exercise 1.13
Problem 5 Exercise 1.14
Problem 6 Exercise 1.18.
(note: For the problems that involve MATLAB printout your data and code including the output directly from the MATLAB environment.)